1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

BRCA1 loss-of-function leads to homologous recombination deficiency (HRD) and increased genomic instability, which elevates somatic mutation rate and predicted neoantigen load (i.e., higher tumor mutational burden, TMB). This mutational landscape enhances tumor immunogenicity and, in the context of PARP inhibitor therapy, drives greater therapeutic vulnerability and potential improvement in progression-free survival. Conversely, BRCA1_reversion_mutations restore HR function, reduce genomic instability, lower TMB/neoantigen load, and diminish sensitivity to PARP inhibitors. These relationships align along the pathway: BRCA1 → HomologousRecombinationRepair → GenomicInstability → TumorMutationalBurden → PARPInhibitorResponse, with network context provided by HR-related genes (e.g., BRCA2, PALB2, RAD51) as optional modifiers to test robustness across HR-pathway perturbations.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: allow assessment of pathway robustness and potential compensatory mechanisms. Feasibility is supported by established assays (DR-GFP for HR, RAD51 foci, γH2AX, micronucleus for instability), sequencing-based measures of TMB/neoantigens, and clinically relevant PARP inhibitor models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions